

Joshua Weidlich

DIZH Bridge Professor for *Digital Higher Education*

Assistant Professor with Tenure Track

University of Zürich /// Zürich University of Teacher Education

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AFFILIATION

University of Zürich

Institut für Erziehungswissenschaft
Kantonsschulstrasse 3, 8001 Zürich
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RESEARCH INTERESTS

Feedback and feedback literacy of students

How students make sense of feedback and act on it, including automated feedback from learning analytics, and how this capacity can be measured and developed.

Learning with generative AI

What AI changes as a source of feedback and a tool for learning, and what the evidence on its effects can and cannot show.

Assessment validity

What assessments still tell us about students' competence when they can use AI.

Research methodology

Causal inference and measurement in educational technology research: which claims a study design can support.

Social presence and collaborative learning

How learners perceive and relate to one another in online and computer-supported group settings.

POSITIONS AND EXPERIENCE

Assistant Professor

2024 – now

University of Zürich & Zürich University of Teacher Education

- Develop research on digital higher education at intersection of UZH & PHZH
- Supervision of doctoral students
- Faculty development

Postdoctoral Researcher

2021 – 2024

Leibniz Institute for Research and Information in Education

- Supervision of PhD students
- Learning Analytics research projects
- Teaching in Computer Science

Scientific staff

2019 – 2021

University of Education, Heidelberg

- Research on Technology in Teacher Education
- Teaching in and development of M.A. “E-Learning and Media Education”

Scientific online tutor 2019 – 2021	FernUniversität in Hagen – Mentoring of B.A. and M.A. theses – Teaching in M.A. module
Scientific staff 2013 – 2018	FernUniversität in Hagen – Research into educational technology in distance education contexts – Teaching online and blended-learning classes – Mentoring B.A. & M.A. theses – Design and development of course materials
Academic Assistant 2017 – 2019	School of Medicine, Heidelberg – Statistical assistance for medical education research projects – Design, development and implementation of a blended learning concept in medical education

EDUCATION

PhD Educational Technology 2015 – 2021	FernUniversität in Hagen Thesis: <i>Social presence in online distance learning environments</i> (abbrev.) Supervisor: Theo J. Bastiaens
M.A. E-Learning and Media Education 2011 – 2013	University of Education, Heidelberg Thesis: <i>The Preparation Phase in Flipped Classrooms – Problems vs Videos</i> (transl.) Supervisors: Christian Spannagel, Holger Meeh
B.A. Educational Science 2007 – 2011	University of Technology, Chemnitz Thesis: <i>Adapting Learning Environments based on Learner's Expertise</i> (transl.) Supervisors: Maria Bannert, Thomas Weiß

HONORS AND AWARDS

2022	Early Career Award for Outstanding Dissertation in Higher Education Awarded by the German Society for Higher Education Teaching (DGHD). Award amount: € 1000
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REFEREED JOURNAL ARTICLES 40

Asterisk (*) denotes mentored student

- Weidlich, J.** (2026). Which inference is at risk? Assessment validity reasoning and generative AI. *Assessment & Evaluation in Higher Education*, 1–20.
- Lawson, A. P., Martella, A. M., **Weidlich, J.**, Mulders, M., & Buchner, J. (2026). Color me confounded: A critical analysis of media comparisons on ChatGPT in education. *Computers & Education*, 254, 105701.
- Kalz, M., Schulze, A., Meeh, H., & **Weidlich, J.** (2026). Validity evidence for the internal structure of the peer feedback orientation scale. *Technology, Knowledge and Learning*, 1–16.
- Weidlich, J.**, & Kalz, M. (2026). How well does teacher education prepare for teaching with technology? A TPACK-based investigation at a university of education. *European Journal of Teacher Education*, 49(1), 168–188.
- Weidlich, J.**, Gotsch, F., Schudel, K., Marusic-Würscher, C., Mazzarella, J., Bolten, H., Bütler, D., Luger, S., Wohlfender, B., & Maag Merki, K. (2025). Teacher, peer, or AI? Comparing effects of feedback sources in higher education. *Computers and Education Open*, 9, 100300.

- Weidlich, J.**, Gašević, D., Drachsler, H., & Kirschner, P. (2025). ChatGPT in education: An effect in search of a cause. *Journal of Computer Assisted Learning*, 41, e70105.
- Xu, K. M., Lin, L., Gorter, M., Schneider, S., **Weidlich, J.**, Davis, R. O., Kreijns, K., & de Groot, R. (2025). Social presence: A key factor in embedding a pedagogical agent into online learning in primary education. *British Journal of Educational Technology*, 1–14.
- Weidlich, J.**, Fink, A., Frey, A., Jivet, I., Gombert, S., Menzel, L., Giorgashvili, T., Yau, J., & Drachsler, H. (2025). Highly informative feedback using learning analytics: How feedback literacy moderates student perceptions of feedback. *International Journal of Educational Technology in Higher Education*, 22(1), 43.
- Richter, E., Spitzer, M. W. H., Morgan, A., Frede, L., **Weidlich, J.**, & Moeller, K. (2025). Large language models outperform humans in identifying neuromyths but show sycophantic behavior in applied contexts. *Trends in Neuroscience and Education*, 39, 100255.
- Weidlich, J.**, Jivet, I., Woitt, S., Orhan Göksün, D., Kraus, J., & Drachsler, H. (2025). The student feedback literacy instrument (SFLI): Multilingual validation and introduction of a short-form version. *Assessment & Evaluation in Higher Education*, 1–17.
- Woitt, S.*, **Weidlich, J.**, Jivet, I., Göksün, D., Drachsler, H., & Kalz, M. (2025). Students' feedback literacy in higher education: An initial scale validation study. *Teaching in Higher Education*, 1–20.
- Weidlich, J.**, Fink, A., Jivet, I., Yau, J., Giorgashvili, T., Drachsler, H., & Frey, A. (2024). Emotional and motivational effects of automated and personalized formative feedback: The role of reference frames. *Journal of Computer Assisted Learning*.
- Kube, D.*, **Weidlich, J.**, Kreijns, K., & Drachsler, H. (2024). Addressing gender in STEM classrooms: The impact of gender bias on women scientists' experiences in higher education careers in Germany. *Education and Information Technologies*, 1–28.
- Kube, D.*, Gombert, S., John, N., **Weidlich, J.**, Kreijns, K., & Drachsler, H. (2024). Escaping binary gender roles: Gender diversity dynamics in a CSCL escape game. *Journal of Computer Assisted Learning*, 1–15.
- Kreijns, K., Yau, J., **Weidlich, J.**, & Weinberger, A. (2024). Towards a comprehensive framework of social presence for online, hybrid, and blended learning. *Frontiers in Education*, 8, 1286594.
- Biedermann, D., Kister, S., Breitwieser, J., **Weidlich, J.**, & Drachsler, H. (2024). Use of digital self-control tools in higher education: A survey study. *Education and Information Technologies*, 29, 9645–9666.
- Kube, D.*, Gombert, S., Suter, B., **Weidlich, J.**, Kreijns, K., & Drachsler, H. (2023). Hacking gender in computer-supported collaborative learning: The experience of being in mixed-gender teams at a computer science hackathon. *Journal of Computer Assisted Learning*, 1–15.
- Weidlich, J.**, Yau, J., & Kreijns, K. (2023). Social presence and psychological distance: A construal level account for online distance learning. *Education and Information Technologies*, 1–23.
- Ferguson, R., Khosravi, H., Kovanović, V., Viberg, O., Aggarwal, A., Brinkhuis, M., ... **Weidlich, J.**, ... & Yan, V. X. (2023). Aligning the goals of learning analytics with its research scholarship: An open peer commentary approach. *Journal of Learning Analytics*, 10(2), 14–50.
- Bozkurt, A., Gjelsvik, T., Adam, T., Asino, T. I., Atenas, J., Bali, M., ... **Weidlich, J.**, & Zawacki-Richter, O. (2023). Openness in education as a praxis: From individual testimonials to collective voices. *Open Praxis*, 15(2), 76–112.
- Weidlich, J.**, Hicks, B., & Drachsler, H. (2023). Causal reasoning with causal graphs in educational technology research. *Educational Technology Research and Development*, 1–19.
- Weidlich, J.**, Gašević, D., & Drachsler, H. (2022). Causal inference and bias in learning analytics: A primer on pitfalls using directed acyclic graphs. *Journal of Learning Analytics*, 9(3), 183–199.
- Kube, D.*, **Weidlich, J.**, Jivet, I., Kreijns, K., & Drachsler, H. (2022). "Gendered differences versus doing gender": A systematic review on the role of gender in CSCL. *Unterrichtswissenschaft*, 1–28.
- Weidlich, J.**, Göksün, D. O., & Kreijns, K. (2022). Extending social presence theory: Social presence divergence and interaction integration in online distance learning. *Journal of Computing in Higher Education*, 1–22.
- Rosar, M.*, & **Weidlich, J.** (2022). Creative students in self-paced online learning environments: An experimental exploration of the interaction of visual design and creativity. *Research and Practice in Technology Enhanced Learning*, 17(8).

- Fominykh, M., **Weidlich, J.**, Kalz, M., & Hybertsen, I. D. (2022). What do they TEL(L)? A systematic analysis of master programs in technology-enhanced learning. *International Journal of Educational Technology in Higher Education*, 19(1), 1–25.
- Weidlich, J.**, Kreijns, K., & Bastiaens, T. (2021). Individual differences in perceptions of social presence: Exploring the role of personality in online distance learning. *Open Education Studies*.
- Ciordas-Hertel, G. P., Rödling, S., Schneider, J., Di Mitri, D., **Weidlich, J.**, & Drachsler, H. (2021). Mobile sensing with smart wearables of the physical context of distance learning students to consider its effects on learning. *Sensors*, 21(19), 6649.
- Wollny, S., Schneider, J., Di Mitri, D., **Weidlich, J.**, Rittberger, M., & Drachsler, H. (2021). Are we there yet? A systematic literature review on chatbots in education. *Frontiers in Artificial Intelligence*, 4, 654924.
- Weidlich, J.**, & Kalz, M. (2021). Exploring predictors of instructional resilience during emergency remote teaching in higher education. *International Journal of Educational Technology in Higher Education*, 18, 41.
- Kreijns, K., Xu, K., & **Weidlich, J.** (2021). Social presence: Conceptualization and measurement. *Educational Psychology Review*, 1–32.
- Baessler, F., **Weidlich, J.**, Schweizer, S., Ciprianidis, A., Bartolovic, M., et al. (2020). What and how are students taught about communicating risks to patients? Analysis of a medical curriculum. *PLOS ONE*, 15(5), e0233682.
- Weidlich, J.**, & Bastiaens, T. J. (2019). Pathway to presence: An investigation of ambient awareness in online learning environments. *Interactive Learning Environments*, 1–14.
- Weidlich, J.**, & Bastiaens, T. J. (2019). Designing sociable online learning environments and enhancing social presence: An affordance enrichment approach. *Computers & Education*, 142, 103622.
- Baessler, F., Zafar, A., Schweizer, S., Ciprianidis, A., Sander, A., Preussler, S., **Weidlich, J.**, ... & Schultz, J.-H. (2019). Are we preparing future doctors to deal with emotionally challenging situations? Analysis of a medical curriculum. *Patient Education and Counseling*, 102(7), 1304–1312.
- Baessler, F., Ciprianidis, A., Rizvi, A. Z., **Weidlich, J.**, Wagner, F. L., Klein, S. B., ... & Schultz, J.-H. (2019). Delirium: Medical students' knowledge and effectiveness of different teaching methods. *The American Journal of Geriatric Psychiatry*, 27(7), 737–744.
- Bovermann, K., **Weidlich, J.**, & Bastiaens, T. J. (2018). Online learning readiness and attitudes towards gaming in gamified online learning: A mixed methods case study. *International Journal of Educational Technology in Higher Education*, 15, 27.
- Weidlich, J.**, & Bastiaens, T. J. (2018). Technology matters: The impact of transactional distance on satisfaction in online distance learning. *International Review of Research in Open and Distributed Learning*, 19(3), 222–242.
- Rosar, M.*, Lipka, A., **Weidlich, J.**, & Bastiaens, T. (2018). Do creative learners prefer inquiry-based learning instructions in digital education programs? *International Journal on E-Learning*, 17(1), 5–16.
- Weidlich, J.**, & Bastiaens, T. J. (2017). Explaining social presence and the quality of online learning with the SIPS model. *Computers in Human Behavior*, 72, 479–487.

PREPRINTS 2

- Weidlich, J.**, Gotsch, F., Schudel, K., Marusic-Würscher, C., Bolten, H., Luger, S., Bütler, D., Wohlfender, B., Mazzarella, J., & Maag Merki, K. (2026). *The medium is not the message: Message features in teacher, peer, and AI feedback help explain the divergence of perceptions and performance*. SocArXiv. https://osf.io/preprints/socarxiv/t5m4k_v1
- Weidlich, J.** (2026). *Do we really care about that? Deep conceptual problems subvert current media comparison practices*. SocArXiv. https://osf.io/preprints/socarxiv/strc7_v1. Presented at the Annual Meeting of the American Educational Research Association (AERA), Los Angeles, USA.

REFEREED PROCEEDINGS 13

Asterisk (*) denotes mentored student

- Menzel-Wuttig, L.*, **Weidlich, J.**, Mendzheritskaya, J., Winter, M., & Drachsler, H. (2026). This is how students think about and act upon personalized, automated feedback on transactivity. In *Mindful TEL: Learning Technologies Shaped with Intention*. EC-TEL 2026. Lecture Notes in Computer Science, vol. 16905 (pp. 420–435). Springer.

- Göksün, D. O., **Weidlich, J.**, Rajagopal, K., Henderikx, M. A., Vrieling-Teunter, E. M., & Kreijns, C. (2025). A control-value account of the association between social presence and satisfaction and perceived learning. In *ISSRIS conference 2025. The 5th International Symposium of Scientific Research and Innovative Studies*.
- Menzel, L.*, Gombert, S., **Weidlich, J.**, Fink, A., Frey, A., & Drachsler, H. (2023). Why you should give your students automatic process feedback on their collaboration: Evidence from a randomized experiment. In O. Viberg, I. Jivet, P. Muñoz-Merino, M. Perifanou, & T. Papathoma (Eds.), *Responsive and Sustainable Educational Futures. EC-TEL 2023. Lecture Notes in Computer Science*, vol. 14200 (pp. 198–212). Springer.
- Ahmad, A., Schneider, J., **Weidlich, J.**, Di Mitri, D., Yau, J. Y. K., Schiffner, D., & Drachsler, H. (2022, April). What indicators can I serve you with? An evaluation of a research-driven learning analytics indicator repository. In *Proceedings of CSEDU conference*.
- Kreijns, K., Henderikx, M., & **Weidlich, J.** (2022, March). Productivity in online CSCL groups: A Rasch analysis approach to the preliminary validation of the productivity scale. In *Proceedings of INTED2022 conference*.
- Bovermann, K., Habla, S., & **Weidlich, J.** (2021, July). Effects of competition in gamified online distance learning on intrinsic motivation: A comparative case study. Association for the Advancement of Computing in Education (AACE).
- Kreijns, K., Henderikx, M., & **Weidlich, J.** (2021, July). Measuring social space in online group learning: Preliminary validation of the social space scale using the Rasch analysis approach. In *EdMedia + Innovate Learning* (pp. 445–458). Association for the Advancement of Computing in Education (AACE).
- Bovermann, K., Habla, S., **Weidlich, J.**, & Bastiaens, T. (2019, June). Better together? A case study comparison of individualistic vs. collectivistic gamification design. In *EdMedia + Innovate Learning* (pp. 1079–1088). Association for the Advancement of Computing in Education (AACE).
- Kreijns, K., **Weidlich, J.**, & Rajagopal, K. (2018). The psychometric properties of a preliminary social presence measure using Rasch analysis. In V. Pammer-Schindler, M. Pérez-Sanagustín, H. Drachsler, R. Elferink, & M. Scheffel (Eds.), *Lifelong Technology-Enhanced Learning. EC-TEL 2018. Lecture Notes in Computer Science*, vol. 11082. Springer.
- Weidlich, J.**, Kreijns, K., Rajagopal, K., & Bastiaens, T. (2018). What social presence is, what it isn't, and how to measure it: A work in progress. In *Proceedings of EdMedia: World Conference on Educational Media and Technology* (pp. 2135–2143). AACE.
- Brunn, C.*, **Weidlich, J.**, & Bastiaens, T. (2018). Effects of questioning advance organizers on learning outcomes in a web-based learning object on research and academic skills. In *Proceedings of EdMedia: World Conference on Educational Media and Technology* (pp. 1969–1978). AACE.
- Gunkel, P.*, **Weidlich, J.**, & Bastiaens, T. (2018). A case study in flipped tutoring: Development, implementation and evaluation of a flipped classroom approach to English language tutoring. In E. Langran & J. Borup (Eds.), *Proceedings of Society for Information Technology & Teacher Education International Conference* (pp. 781–786). AACE.
- Weidlich, J.**, & Spannagel, C. (2014). Die Vorbereitungsphase im Flipped Classroom. Vorlesungsvideos versus Aufgaben. In K. Rummler (Hrsg.), *Lernräume gestalten – Bildungskontexte vielfältig denken* (S. 237–248). Waxmann.

EDITED VOLUMES 1

- Weidlich, J.**, Chounta, I.-A., Tammets, K., Nazaretsky, T., Limbu, B., Wild, F., & Arnau-González, P. (Eds.). (2026). *Mindful TEL: Learning Technologies Shaped with Intention. 21st European Conference on Technology Enhanced Learning, EC-TEL 2026, Proceedings, Part I. Lecture Notes in Computer Science*, vol. 16905. Springer.

BOOKS AND BOOK CHAPTERS 3

- Wöhrle, J., Karolyi, H., **Weidlich, J.**, Seidenberg, N., Blanc, B., Woick, A., Drachsler, H., Pinkwart, N., Markgraf, D., & de Witt, C. (2025). *LAMASS-Studie: Studienerfolg und Studienabbruch in digitalen Studienformaten*. FernUniversität in Hagen.
- Kreijns, K., **Weidlich, J.**, & Kirschner, P. (2024). Identifying the pitfalls of social interaction in online computer-supported group learning: A revisit and update. In *The Cambridge Handbook of Cyber Behavior*. Cambridge University Press.

Kreijns, K., Bijker, M., & **Weidlich, J.** (2020). A Rasch analysis approach to the development and validation of a social presence measure. In *Rasch Measurement* (pp. 197–221). Springer.

INVITED TALKS AND PRESENTATIONS 11

Weidlich, J. (2025, June). Keynote: *Principles and Pitfalls of Designing Effective Feedback*. Universität Zürich (Abschlusscolloquium CAS Hochschuldidaktik).

Weidlich, J. (2025, May). Keynote: *Smart Education requires Mindful Feedback*. Digital Society Initiative (DSI Community Education Conference).

Weidlich, J. (2025, April). Invited panel member: *UZHai – Akzeptanz und Folgen von AI in Studium und Lehre*. Universität Zürich (Ständetalk 2025).

Weidlich, J. (2025, April). *Closing the Loop mit Feedback?* Presented at Universität Stuttgart. Presented at Universität Graz (QM Netzwerktagung).

Weidlich, J. (2025, March). Invited panel member: *Learning Analytics 2035: Pushing the Boundaries and Meeting the Challenges*. University of Dublin (Learning Analytics Conference).

Weidlich, J. (2025, February). *Digital verbunden: Erkenntnisse und Perspektiven zu Social Presence in Online-Lernumgebungen*. Presented at Universität Stuttgart (Präsenz gestalten).

Weidlich, J. (2023, November). Invited as discussant on digitalization in higher education. Symposium *Approaches to Research in Higher Education Teaching and Learning* at Pädagogische Hochschule Zürich.

Weidlich, J. (2021, May). *Die Bedeutung sozialer Präsenz in der coronabedingten Online-Lehre* [transl.: The relevance of social presence in emergency remote teaching]. Presented online, Hochschule für Angewandte Wissenschaften Hamburg.

Weidlich, J. (2020, January). *Recent findings in social presence research*. Presented at the inaugural symposium of Karel Kreijns at Open University of the Netherlands.

Weidlich, J. (2018, November). *Social affordances, social presence and sociable online learning environments: A study of the SIPS model in action*. Presented online at CIDER Sessions, Athabasca University.

Weidlich, J. (2018, January). *Enhancing social presence in online learning through the use of social software*. Presented at Heidelberg University of Education.